

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Mid-Semester Examination, March, 2019

First (1st) Year

Agric, Chemical, Mechanical, Material and Computer Engineering

COE/EE 152: BASIC ELECTRONICS

TIME: 60 MINS

PROGRAM OF STUDY:

INDEX NO:.....

Instructions

- i. Candidates are to indicate their Index Number and Program of Study in the spaces provided.*
- ii. Candidates are to submit ONLY the Scannable Sheet.*
- iii. Choose from the options lettered A to D, the most suitable answer to the question statements*
- iv. Shade the most suitable answer to the question statements on the Scannable Sheet provided*
- v. Candidates are to write the answer at the back of the Scannable Sheet if he/she thinks the right answer cannot be found in the options provided*
- vi. Each correct answer carries 0.5 a mark*

**CANDIDATES ARE ALLOWED TO TAKE THE QUESTION PAPERS
AWAY**

MULTIPLE CHOICE. Choose the **one** alternative that best completes the statement or answers the question.

- 1) What can be said regarding the energy level of an electron?
 - A) Electrons in the shell closest to the nucleus of an atom have the highest energy level
 - B) Electrons in the outermost shell have the highest energy level
 - C) All electrons have the same energy level
 - D) Electrons exhibit energy characteristics only in the presence of protons
- 2) A linear component is one in which an increase in current is proportional to the applied voltage as given by Ohm's law. Which of the following is an example of a linear component?
 - A) Resistor
 - B) Capacitor
 - C) Diode
 - D) Transistor
- 3) To make a p-type of semiconductor material you need a doping material that is _____.
 - A) trivalent
 - B) tetravalent
 - C) pentavalent
 - D) hexavalent
- 4) If one silicon diode and one germanium diode are connected in series, the voltage drop across the combination of the two diodes will be equal to _____.
 - A) the forward drop equal to that of the germanium diode
 - B) the forward drop equal to that of the sum of the voltage drops across the two diodes
 - C) the forward drop equal to that of the difference of the voltage drops across the two diodes
 - D) the forward drop equal to that of the silicon diode
- 5) The point of intersection between the characteristic curve of the diode and the resistors loadline is known as the _____.
 - A) Q-point
 - B) point of operation
 - C) quiescent point
 - D) All of the above
- 6) The direction of the arrow in the diode symbol points in the direction of _____.
 - A) leakage current flow
 - B) from n-type of semiconductor to p-type semiconductor material
 - C) from p-type of semiconductor to n-type semiconductor material
 - D) positive terminal under forward bias
- 7) The reverse saturation current of a diode will just about _____ for every 10°C rise in the diode temperature.
 - A) decrease proportionately with temperature
 - B) half
 - C) double
 - D) increase proportionately with temperature
- 8) Increasing the temperature of a forward-biased diode _____.
 - A) causes forward current to decrease
 - B) has no significant effect on the forward current
 - C) causes forward current to increase
 - D) None of these
- 9) The piecewise linear model, equivalent circuit of the diode consists of _____.
 - A) a battery and the ideal diode
 - B) a junction capacitor, a battery, a small resistor, and the ideal diode
 - C) a battery, a small resistor, and the ideal diode
 - D) the ideal diode

10) Some of the modern ohmmeters have a diode test setting. If you do not have one of these ohmmeters then to test the diode you need to check its resistance in the forward and the reverse direction. These resistances should be _____.

- A) relatively low in the forward direction and relatively low in the reverse direction
- B) relatively high in the forward direction and relatively high in the reverse direction
- C) relatively high in the forward direction and relatively low in the reverse direction
- D) relatively low in the forward direction and relatively high in the reverse direction

11) In the Zener region the current _____ and the voltage across the diode _____.

- A) can increase a lot; can increase a lot
- B) can increase a lot; is almost constant
- C) is almost constant; is almost constant
- D) is almost constant; can increase a lot

12) Light-emitting diodes emit light when the p-n junction is _____.

- A) zero biased
- B) operating in the Zener region
- C) forward-biased
- D) reverse-biased

13) As semiconductor devices have become _____ one of the primary purposes of the container is simply to provide a means for physical handling.

- A) widely used
- B) miniaturized
- C) more powerful
- D) larger

14) An advantage of the miniaturization of electronic devices is that they _____.

- A) increase availability
- B) reduce cost
- C) increase speed
- D) improve reliability

15) The _____ diode is a short circuit for the region of conduction and it is an open circuit in the region of non-conduction.

- A) power
- B) typical
- C) small-signal
- D) ideal

16) The ideal diode symbol has an arrow that points in the direction of _____.

- A) the forward current flow
- B) positive terminal under forward bias
- C) the leakage current flow
- D) All of the above

17) The term _____ is applied to any material that supports a generous flow of charge when a voltage source of limited magnitude is applied across its terminals.

- A) conductor
- B) insulator
- C) dielectric
- D) semiconductor

18) The term _____ is applied to a material that offers a very low level of conductivity under pressure from an applied voltage.

- A) conductor
- B) semiconductor
- C) insulator
- D) ionic

19) The term _____ is applied to a material that has a conductivity level somewhere between the extremes of conductivity.

- A) semiconductor
- B) insulator
- C) conductor
- D) ionic

20) Which of the following is not a commonly used semiconductor material?

- A) germanium
- B) carbon
- C) silicon
- D) lead

- 21) As the device temperature increases, semiconductor materials tend to have _____.
 A) an increasing number of free electrons
 B) a decreasing number of free electrons
 C) relatively unchanged conduction levels
 D) lower conduction levels
- 22) Pentavalent elements have _____ valence electrons.
 A) 5 B) 4 C) 3 D) 1
- 23) Doping is used to _____.
 A) stabilize the conductivity of an intrinsic semiconductor
 B) decrease the conductivity of an intrinsic semiconductor
 C) increase the conductivity of an intrinsic semiconductor
 D) increase the insulative quality of an intrinsic semiconductor
- 24) When pentavalent elements are used in doping, the resulting material is called _____ material and has an excess of _____.
 A) n-type; valence-band holes B) n-type; conduction-band electrons
 C) p-type; conduction-band electrons D) p-type; valence-band holes
- 25) When trivalent elements are used in doping, the resulting material is called _____ material and has an excess of _____.
 A) p-type; valence-band holes B) n-type; valence-band holes
 C) n-type; conduction-band electrons D) p-type; conduction-band electrons
- 26) In an n-type material, the majority carriers are _____.
 A) valence-band electron B) conduction-band electrons
 C) conduction-band holes D) valence-band holes
- 27) When a p-n junction's depletion layer is narrowed and the device acts as a nearly perfect conductor, it is _____.
 A) forward-biased B) unbiased C) reverse-biased D) None of the above
- 28) The maximum reverse bias potential that can be applied to a Zener diode before it enters the Zener region is called the _____.
 A) PIV B) threshold voltage C) depletion voltage D) barrier voltage
- 29) When a p-n junction is reverse-biased, the depletion layer is _____ and the device acts as a near-perfect _____.
 A) narrowed; conductor B) widened; conductor
 C) narrowed; insulator D) widened; insulator
- 30) The diode electrode with n-type material is called the _____.
 A) Zener region B) depletion region C) anode D) cathode
- 31) Silicon diodes have been more significantly developed than germanium because
 A) it is more tolerant of heat B) it is cheaper
 C) it has a lower forward voltage drop D) it is easier to produce

- 32) In a p-type material, the minority carriers are _____.
 A) valence-band holes B) valence-band electrons
 C) conduction-band electrons D) conduction-band electrons
- 33) Pentavalent atoms are often referred to as _____.
 A) donor atoms B) majority carriers C) acceptor atoms D) minority carriers
- 34) When a p-n junction is reverse-biased, its junction resistance is _____.
 A) high B) determined by the components that are external to the device
 C) constantly changing D) low
- 35) A p-n junction is forward biased when _____.
 A) the applied potential causes the n-type material to be more positive than the p-type material
 B) the applied potential causes the n-type material to be more negative than the p-type material
 C) both materials are at the same potential
 D) None of these
- 36) A p-n junction is reverse biased when _____.
 A) the applied potential causes the p-type material to be more negative than the n-type material
 B) the applied potential causes the n-type material to be more positive than the p-type material
 C) the current flow across the junction is based on minority carrier transfer
 D) All of the above
- 37) The isolated atomic energy structure associated with electron orbital shells is called a/an _____.
 A) conduction band B) energy gap C) energy band D) valence band
- 38) The diode electrode with p-type material is called the _____.
 A) Zener region B) anode C) cathode D) depletion region
- 39) The diffusion capacitance of a diode is a shunt capacitance effect that occurs when the diode.....
 A) is small B) is reverse biased C) is large D) is forward biased
- 40) The transition capacitance of a diode is a shunt capacitive effect that occurs when the diode
 A) is large B) is reverse-biased C) is small D) is forward-biased
- 41) Name the two most common semiconductor materials.
 A) Silicon and germanium B) Germanium and carbon
 C) Gallium and Arsenide D) Gallium and silicon
- 42) Doping is the process of adding impurities to pure semiconductor material. Impurities define the majority and minority carriers in a semiconductor. Which of the following statements is true?
 A) Trivalent impurities cause electrons to be the majority carriers
 B) Pentavalent impurities cause holes to be the majority carriers
 C) Pentavalent impurities add one extra hole to the covalent bond
 D) Trivalent impurities cause holes to be the majority carriers
- 43) Which one of the following devices is roughly equivalent to a diode?
 A) A one-way valve B) A pressure regulator
 C) A spring D) None of the above

- 44) What can be said about the depletion region in a diode?
- A) The depletion region size is independent of barrier voltage
 - B) The size of the depletion region is inversely proportional to barrier potential
 - C) The depletion region shrinks as the diode is reverse biased
 - D) The depletion region acts as an insulator and the dielectric value is proportional to barrier potential
- 45) Which best describes a conducting forward-biased diode?
- A) Negative cathode, positive anode $V_F = 0.2$ volts
 - B) Positive cathode, negative anode $V_F = 0.2$ volts
 - C) Positive cathode, negative anode $V_F = 10$ volts
 - D) Negative cathode, positive anode $V_F = 0.7$ volts
- 46) What denotes the maximum voltage a diode can withstand when reverse-biased?
- A) Peak inverse voltage
 - B) Reverse breakdown
 - C) Reverse bias
 - D) Peak forward-bias voltage
- 47) If the forward bias is less than the barrier potential, what condition is the diode in?
- A) The diode will actually conduct some small amount of current
 - B) The diode will not conduct until forward bias is greater than the barrier potential
 - C) The diode will function as a 1k resistor
 - D) The diode will conduct heavily because the barrier potential is greater than the forward bias
- 48) What is the purpose of a current-limiting resistor in a diode circuit?
- A) The resistor prevents the diode from becoming reverse-biased
 - B) A properly biased diode is like a short circuit, therefore a resistor provides a voltage drop
 - C) The resistor will increase the PIV rating of the diode
 - D) A current-limiting resistor is only necessary if it is reverse biased
- 49) Using an ohmmeter to check a diode, what indicates the diode is likely good?
- A) High resistance measured regardless of how the leads are connected
 - B) Low resistance measured only with positive lead on cathode and negative lead on anode
 - C) Low resistance measured only with positive lead on anode and negative lead on cathode
 - D) Low resistance regardless of how the leads are connected
- 50) In a conductor, the energy levels of the conduction and valence band
- A) are very close in value
 - B) overlap
 - C) are negative
 - D) are the same
- 51) When recombination occurs, an electron
- A) drops into the valence band
 - B) drops into a hole
 - C) loses energy
 - D) all of the above
- 52) The majority carriers in the p region of a diode
- A) are much less in number than the majority carriers in the n region
 - B) are much fewer in number than the majority carriers in the n region
 - C) are electrons
 - D) are holes

The diodes used in a bridge rectifier circuit have the following parameters:

$$V_T = 0V, \quad r_f = 15\Omega, \quad R_R = \infty, \quad V_Z = \text{very large}$$

The secondary winding of the transformer has a resistance $R_S = 0\Omega$. A load of 970Ω is connected across its output. If the voltage across the secondary is $v_s = 25\sin wt$ volts,

Use the above information to solve questions 53 to 60

53) Determine the PIV rating of the diodes

- a. 30V
- b. 20V
- c. 25V
- d. 10V
- e. None of the above

54) Determine the DC current through the load considering the voltage drop across the Diodes

- a. 0.06524A
- b. 0.14720A
- c. 0.01472A
- d. 0.01592A
- e. None of the above

55) Determine the DC Voltage on the load considering the voltage drop across the Diodes

- a. 15.428V
- b. 15.676V
- c. 15.563V
- d. 15.378V
- e. None of the above

56) Determine the DC power supplied to the load

- a. 0.2600W
- b. 0.2400W
- c. 0.2300W
- d. 0.2500W
- e. None of the above

57) What is the conversion efficiency of this rectifier

- a. 0.7737pu
- b. 0.7685pu
- c. 0.7857pu
- d. 0.7537pu
- e. None of the above

58) What is the peak value of the AC current?

- a. 0.02500A
- b. 0.02300A
- c. 0.02400A
- d. 0.02200A
- e. None of the above

59) What is the RMS value of the current supplied to the circuit?

- a. 0.018020A
- b. 0.016128A
- c. 0.017680A
- d. 0.015040A
- e. None of the above

60) What is the AC power supplied to the whole circuit?

- a. 0.31428A
- b. 0.31267A
- c. 0.32428A
- d. 0.32040A
- e. None of the above